

Step-by-Step Working Process

mermaid

flowchart TD

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A["Step 1: Input (User Prompt ya Visual Canvas)"] --> B["Step 2: AI Engine Analysis & Node Placement"]
B --> C["Step 3: Real-Time Cost & AI Doctor Audit"]
C --> D["Step 4: Multi-Cloud IaC Code Compilation"]
D --> E["Step 5: DevOps Execution (Plan, Apply & Simulation)"]
E --> F["Final Outputs (Terraform, Docker, K8s, Architecture)"]
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Step 1: User Input & Requirement Gathering

Aapke paas do tarike hote hain:

1. **Conversational AI Prompt:** Aap AI Chat me apni requirement likhte hain (jaise: *"Mujhe 50k users ke liye food delivery app ka architecture chahiye with PostgreSQL and Redis"*).
2. **Interactive Visual Canvas:** Aap canvas par directly cloud components (ALB, EC2, RDS, Redis, S3, CloudFront, etc.) drag-and-drop karke connect kar sakte hain.

Step 2: AI Reasoning & Canvas Synchronization

- Background me AI Engine (Groq LLaMA / Qwen / Gemini) aapke prompt ko analyze karta hai.
- Ye best cloud design patterns (VPC, Public/Private Subnets, Multi-AZ High Availability) select karta hai.
- Visual Canvas par automatically nodes aur data flow arrows generate ho jaate hain.

Step 3: Cost Intelligence & AI Doctor Audit

Jaise hi architecture canvas par banta hai, system automatically do checks karta hai:

1. **Real-time Cost Estimation:** Har resource (EC2 instance type, RDS storage, Redis cluster, NAT Gateway) ke AWS/GCP pricing ke mutabik **Monthly Cloud Cost (\$/month)** calculate karta hai.
2. **AI Doctor (Security & Well-Architected Audit):** Architecture me faults detect karta hai:

Step 4: Automated IaC Code Generation

Aapke canvas par maujood components ko read karke `server/iacGenerator.js` teen tarah ke production-ready codes bina kisi human error ke build karta hai:

1. **Terraform (`main.tf`):**
 - Provider setup, VPC, Route 53 Private/Public DNS.
 - EC2 Auto Scaling fleet with automated user bootstrap script.
 - Multi-AZ RDS PostgreSQL with **AWS Secrets Manager** (koi hardcoded password nahi).
 - Multi-AZ High Availability **Redis Replication Group**.
 - CloudFront CDN + AWS WAFv2 Layer 7 DDoS/Rate-limit Protection.

- CloudWatch Automated Metric Alarms (CPU > 80%, ALB 5XX errors, Storage < 5GB).

2. **Docker Compose** (`docker-compose.yml`): Local machine par ek click me pura stack run karne ke liye.
3. **Kubernetes Manifests** (`k8s-manifest.yaml`): Containerized environment me cluster orchestration ke liye.

Step 5: DevOps Execution & Live Simulation

- **Terraform Plan:** Code execute karne se pehle resources ka dependency graph validate karta hai.
- **Deploy to Cloud (Apply):** Cloud credentials ke sath live AWS/GCP account par actual resources provision karta hai.
- **Docker Stack:** Local development environment spawn kar deta hai.